Birla Institute of Technology & Science, Pilani Hyderabad Campus
Computer Science and Information Systems Department
Comprehensive Exam – I Semester – 2023-24

CS F407 Artificial Intelligence

PART - B

SCHEME OF EVALUATION

Max Marks: 90
Part A – 25M
Part B – 65M

1 **BLIND AND INFORMED SEARCH** [5M]

A* states along with the costs using OPEN & CLOSED **[2 marks]**
UCS states along with the costs using OPEN & CLOSED **[2 marks]**
Justify the differences: More states explored in UCS – so inefficient **[1 mark]**

A*	
OPEN	**CLOSED**
S-11	S-11
A-13, D-12	D-12
A-13, E-12,	E-12
A-13, B-15, F-12	F-12
A-13, B-15, G-12	G-12
A-13, B-15	G-12

UCS	
S	S
A-3, D-3	D-3
A-3, E-5	A-3
E-5, B-7,	E-5
B-7, F-9	B-7
F-9, C-11	F-9
C-11, G-12	C-11
G-12,	G-12

2 **UNSUPERVISED LEARNING**

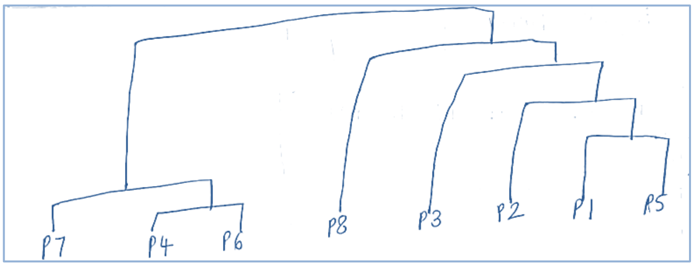
Construct Dendrograms using single and complete link approaches and comment on the differences in the outcomes.
Euclidean metric for computing distance.
If all values correct – 4 marks; partial [2 marks]

Vis	Vol		S1	S2	S3	S4	S5	S6	S7	S8
0.62	0.27	**S1**	0	0.3162	0.2941	0.8169	0.2062	0.7965	0.813	0.4776
0.88	0.45	**S2**	0.3162	0	0.6103	0.9005	0.211	0.8782	0.858	0.7103
0.38	0.1	**S3**	0.2941	0.6103	0	0.8476	0.4701	0.8317	0.8807	0.3912
0.1	0.9	**S4**	0.8169	0.9005	0.8476	0	0.714	0.0224	0.1082	0.4752
0.67	0.47	**S5**	0.2062	0.211	0.4701	0.714	0	0.692	0.6859	0.5016
0.12	0.89	**S6**	0.7965	0.8782	0.8317	0.0224	0.692	0	0.099	0.4627
0.19	0.96	**S7**	0.813	0.858	0.8807	0.1082	0.6859	0.099	0	0.5304
0.17	0.43	**S8**	0.4776	0.7103	0.3912	0.4752	0.5016	0.4627	0.5304	0

All dendrogram stages/matrices must be shown

Single link: **Each stage** of dendrogram carries mark as shown below.

1	(s4, s6)	
2	**(s7, s4, s6)**	1 mark
3	**(s1, s5) or (s2, s5)**	1 mark
4	(s2, s1, s5)	
5	(s2, s1, s5, s3)	
6	**(s3, s2, s1, s5, s8)**	1 mark
7	**(s7, s4, s6, s3, s2, s1, s5, s8)**	



Fully correct dendrogram diagram – 2 marks

	S1	S2	S3	S5	S8	S7, S4, S6
S1	0	0.32	0.3	0.21	0.48	0.8
S2		0	0.61	0.21	0.71	0.86
S3			0	0.47	0.39	0.83
S5				0	0.5	0.69
S8					0	0.46
S7, S4, S6						0

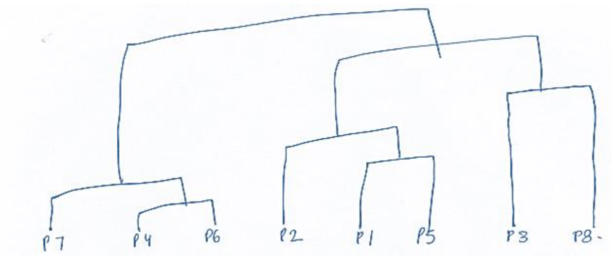
	S2	S3	S8	S7, S4, S6	S1, S5
S2	0	0.61	0.71	0.86	0.21
S3		0	0.39	0.83	0.3
S8			0	0.46	0.48
S7, S4, S6				0	0.67
S1, S5					0

	S3	S8	S7, S4, S6	S2, S1, S5
S3	0	0.39	0.83	0.3
S8		0	0.46	0.48
S7, S4, S6			0	0.69
S2, S1, S5				0

	S8	S7, S4, S6	S3, S2, S1, S5
S8	0	0.46	0.39
S7, S4, S6		0	0.69
S3, S2, S1, S5			0

Complete link:

1	(s4, s6)	
2	(s7, s4, s6)	
3	(s1, s5) or (s2, s5)	1 mark
4	(s2, s1, s5)	
5	**(s3, s8)**	2 mark
6	(s2, s1, s5, s3, s8)	
7	**(s7, s4, s6, s3, s2, s1, s5, s8)**	

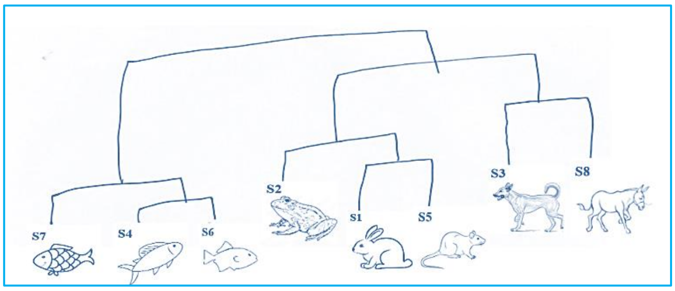


Fully correct dendrogram diagram – 2 marks

Similar matrices must be shown

	S3, S8	S7, S4, S6	S2, S1, S5
S3, S8	0	0.9	0.88
S7, S4, S6		0	0.71
S2, S1, S5			0

Comment on the differences: Complete link yields more relevant clusters at different levels in the dendrogram!!!! [1 mark]



3	

SUPERVISED LEARNING

Train a **Bayesian Classifier** using the given training instances t_1-t_{14}.

Test the Classifier with the three Testing (Query) Instances.

Note: Computation of prior and conditional probabilities must be shown.

Computing **prior probability** of each class: **(1 mark)**

CLASS	N=14 (t_1-t_{14})	**prior probability**
+ve	9/14	0.6428
-ve	5/14	0.3571

Compute the required **conditional probabilities** as following: **(8 marks)**

A	**+ve**	**-ve**
a1	0.2222	0.6
a2	0.3333	0.4

B	**+ve**	**-ve**
b2	0.4444	0.4

C		**+ve**	**-ve**
c1		0.6667	0.2
c2		0.3333	0.8

D	**+ve**	**-ve**
d1	0.3333	0.6
d2	0.6667	0.4

Using these probabilities, obtain P($Query_i$| C_i)P(C_i);

predicted class label is the class C_i for which this expression is maximum. **(2+2+2 marks)**

Query 1: P(Query1|+ve) → P(A= a1|Class = +ve) × P(B= b2|Class = +ve)
× P(C= c1|Class = +ve) × P(D= d2|Class = +ve)
= 0.222 × 0.444 × 0.667 × 0.667 = 0.0438

P(Query1|-ve) → P(A= a1|Class = -ve) × P(B= b2|Class = -ve)
× P(C= c1|Class = -ve) × P(D= d2|Class = -ve)
= 0.6 × 0.4 × 0.2 × 0.4 = 0.0192

To find the class, C_i , that maximizes P(Query1| C_i)P(C_i), we compute

P(Query1|+ve)P(Class = +ve) = 0.044×0.643 = 0.0282

P(Query1|-ve)P(Class = -ve) = 0.019×0.357 = 0.0068

The Bayesian classifier predicts **CLASS = +ve** for **Query1.**

Query 2: P(Query2|+ve) → P(A= a1|Class = +ve) × P(B= b2|Class = +ve)
× P(C= c2|Class = +ve) × P(D= d1|Class = +ve)
= **0.222 × 0.444 × 0.333 × 0.333 = 0.0109**

P(Query2|-ve) → P(A= a1|Class = -ve) × P(B= b2|Class = -ve)
× P(C= c2|Class = -ve) × P(D= d1|Class = -ve)
= **0.6 × 0.4 × 0.8 × 0.6 = 0.1152**

To find the class, C_i , that maximizes P(Query2| C_i)P(C_i), we compute

P(Query2|+ve)P(Class = +ve) = 0.0109×0.643 = 0.0070

P(Query2|-ve)P(Class = -ve) = 0.1152×0.357 = 0.0411

The Bayesian classifier predicts **CLASS = -ve** for ***Query2.***

Query 3: P(Query3|+ve) → P(A= a2|Class = +ve) × P(B= b2|Class = +ve)
× P(C= c1|Class = +ve) × P(D= d2|Class = +ve)
= **0.333× 0.444 × 0.666 × 0.666 = 0.0658**

P(Query3|-ve) → P(A= a2|Class = -ve) × P(B= b2|Class = -ve)
× P(C= c1|Class = -ve) × P(D= d2|Class = -ve)
= **0.4 × 0.4 × 0.2 × 0.4 = 0.0128**

To find the class, C_i , that maximizes P(Query1| C_i)P(C_i), we compute

P(Query3|+ve)P(Class = +ve) = 0.0658×0.643 = 0.0432

P(Query3|-ve)P(Class = -ve) = 0.0128×0.357 = 0.0045

The Bayesian classifier predicts **CLASS = +ve** for ***Query3.***

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4 **KNOWLEDGE REPRESENTAION & REASONING**

Consider a FIS used for clinical Diagnosis. The system is meant to diagnose the severity of a disease as mild, moderate or severe on a scale of 10.
The outcome (severity) depends on the inputs, body temperature and headache intensity. All input and output variables are scaled to a range of 0 to 10. The triangular membership functions (trimf) and trapezoidal membership functions(trapmf) are used for characterizing the fuzzy sets in all the variables.

The trimf and trapmf values for the variables are defined as follows:
Input 1- *temperature*: $\text{trapmf}_{\text{LOW}}$ = a=0, b=0, c=3, d=5; $\text{trimf}_{\text{MEDIUM}}$ = a=3, b=5, c=7;
$\text{trapmf}_{\text{HIGH}}$ = a=5, b=7, c=10, d=10;
Input 2- *headache*: $\text{trapmf}_{\text{MODERATE}}$ = a=0, b=1, c=4, d=6; $\text{trimf}_{\text{SEVERE}}$ = a=4, b=6, c=8;
$\text{trapmf}_{\text{EXTREME}}$ = a=6, b=8, c=10, d=10;
Output – *severity*: $\text{trapmf}_{\text{MILD}}$ = a=0, b=0, c=3, d=5; $\text{trimf}_{\text{MODERATE}}$ = a=3, b=5, c=7;
$\text{trapmf}_{\text{SEVERE}}$ = a=5, b=7, c=10, d=10;

The inference rules for the fuzzy system are defined as follows:
RULE 1: IF *temperature* = high AND *headache* = moderate THEN severity = mild;
RULE 2: IF *temperature* = high AND *headache* = extreme THEN severity = severe;
RULE 3: IF *temperature* = moderate THEN severity = mild;
RULE 4: IF *temperature* = moderate OR *headache* = severe THEN severity = moderate;

4i **FUZZIFICATION**: Fuzzify the input variables for the following values: *temperature* = 4; *headache* = 7; [4M]

Fuzzification of Temperature: [2 marks]
$\mu_{trapmfLOW}(4) = 0.5;$ $\mu_{trimfMEDIUM}(4) = 0.5;$ $\mu_{trapmfHIGH}(4) = 0;$

Fuzzification of Headache Intensity: [2 marks]
$\mu_{trapmfMODERATE}(7) = 0;$ $\mu_{trimfSEVERE}(7) = 0.5;$ $\mu_{trapmfEXTREME}(7) = 0.5;$

IMPLICATION: [6M]

4ii
RULE 1: IF *temperature* = high **AND** *headache* = moderate THEN severity = mild; [2 marks]
$\mu_{trapmfHIGH}(4) = 0$ $\mu_{trapmfMODERATE}(7) = 0;$ $\mu_{trapmfMILD}() = 0$

RULE 2: IF *temperature* = high **AND** *headache* = extreme THEN severity = severe; [2 marks]
$\mu_{trapmfHIGH}(4) = 0$ $\mu_{trapmfEXTREME}(7) = 0.5;$ $\mu_{trapmfSEVERE}() = 0$

RULE 3: IF *temperature* = moderate **// ** wrong rule; temp moderate does not exist***//** [2 marks]

RULE 4: IF *temperature*** = moderate **OR** *headache* = severe THEN severity = moderate;
$\mu_{trimfMEDIUM}(4) = 0.5;$ $\mu_{trimfSEVERE}(7) = 0.5;$ $\mu_{trimfMODERATE}() = 0.5$

5 **INTELLIGENT SYSTEMS & ARCHITECTURE**

5i What is a CBR System? List it's phases [3M]
CASE- BASED REASONING (CBR) **[1 mark]**
Case-based reasoning (CBR) uses an explicit database of problem solutions to address new problem-solving situations.
These solutions may be collected from human experts through the knowledge engineering process or may reflect the results of previous search-based successes or failures.

RETRIEVE – REUSE – RETAIN – REVISE **[2marks]**

5ii How you will use a CBR system to handle problems in customer service; State and explain the methods and metrics that you will use (if any) at different phases. [7M]

RETRIEVE **[2marks]**

The retrieval task is defined as finding a small number of cases from the case-base with the highest similarity to the query. [5M]

This is a k-nearest-neighbor retrieval task considering a specific similarity function **[1 mark]**.

REUSE **[2marks]**

Reusing a retrieved solution can be quite simple if the solution is returned unchanged as the proposed solution for the new problem.

OR

REVISE

Revising a retrieved solution with minor changes

RETAIN **[2marks]**

The retain phase/feedback is the learning phase of a CBR system (adding a revised case to the case base).

• Explicit competence models have been developed that enable the selective retention of cases (because of the continuous increase of the case-base).

5iii Model a problem/case in the customer servise scenario; Identify the root cause, factors, and therby schematically represent it as a AOR graph.

Note: Descriptive explanation not required

Draw an **AND-OR Graph [5 marks]** for any problem: Eg. Order not on time

6 Design a **single machine learning pipeline:** [5M]

Train vs Test [1 mark].

fuzzy logic for uncertainty handling, **[1 mark]**.

features extracted from images, **[1 mark]**.

cooperative search for feature subset selection **[1 mark**

discriminant learner for supervised learning. **[1 mark]**